EXPT NO:4 **DATE: 24.07.2026**	**TRANSFORM DATA TO SUIT BI REPORTING REQUIREMENTS AND DESIGN A DASHBOARD WITH KEY PERFORMANCE INDICATORS (KPIs) USING TABLEAU AND POWERBI**

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1.What is the role of data transformation in BI reporting?
Data transformation plays a critical role in Business Intelligence reporting because it is the process of converting raw, unstructured data into a clean and usable format. When businesses pull data from various sources, it is rarely ready for immediate analysis, often containing mismatched formats or poorly organized text. Transforming this data involves restructuring it, changing data types, and creating new calculated columns so that it perfectly aligns with the specific reporting requirements. By performing these operations, analysts ensure that the underlying data model is robust, logical, and connected correctly. Ultimately, this foundational step guarantees that the final reports and dashboards generated are accurate, consistent, and capable of providing reliable insights for high-level decision-making.

2,What are the common data cleaning techniques in Tableau and Power BI?
Common data cleaning techniques in both Tableau and Power BI involve several key steps to prepare raw datasets for accurate visualization. One primary technique is the removal of duplicate records and the handling of missing or null values, which can be done by filtering them out or replacing them with default zero values. Another standard procedure is splitting columns, where a single text field containing multiple pieces of information is divided into separate, distinct columns for easier sorting. Analysts also frequently change data types to ensure numbers, dates, and text are recognized correctly by the calculation engines of the software. Additionally, renaming columns for clarity and trimming excess spaces from text fields are standard practices that help maintain a pristine and highly readable dataset before any visual analysis begins.

3.How do Tableau and Power BI handle data aggregation?
Tableau and Power BI handle data aggregation by automatically summarizing granular data points into higher-level metrics based on the dimensions present in the visualization. When a user drags a numerical field into a view, both tools intuitively apply a default aggregation function, most commonly a sum or a count, to present the data meaningfully without overwhelming the screen. Users can easily change these default settings from a dropdown menu to calculate averages, minimums, maximums, or variances depending on the specific analytical need. Power BI often utilizes Data Analysis Expressions to create highly complex and custom aggregated measures that dynamically respond to user slicers and filters. Similarly, Tableau uses level of detail expressions to allow analysts to control the exact granularity of the aggregation, ensuring accurate calculations regardless of how the visual layout is structured.

4.What are KPIs, and why are they important in a dashboard?
Key Performance Indicators are specific, quantifiable measurements used to gauge a company's overall long-term performance or the success of a particular operational activity. They are incredibly important in a dashboard because they provide an immediate, high-level snapshot of business health without requiring the user to interpret complex charts or dig through dense tables of numbers. By placing these metrics prominently at the top of a report, stakeholders can instantly see if they are meeting their strategic targets, such as revenue goals, error margins, or customer retention rates. They essentially act as a compass for the organization, highlighting

areas that are performing well and flagging those that require urgent attention. Without clear indicators, a dashboard risks becoming a confusing collection of data rather than a focused tool that drives strategic business decisions.

5.What are the different types of charts available in Tableau and Power BI for visualization?

Both Tableau and Power BI offer a vast array of chart types to cater to diverse analytical requirements and different underlying data structures. For tracking changes over time, line charts and area charts are standard offerings that clearly illustrate continuous trends and historical movements. When comparing distinct categories, users frequently rely on bar charts, column charts, and tree maps to show relative sizes and part-to-whole relationships across a dataset. For spatial and geographical analysis, both platforms provide robust mapping capabilities, including filled regional maps and coordinate-based symbol maps. Furthermore, they support specialized visualizations like scatter plots for identifying correlations between two variables, gauge charts for tracking progress against a specific goal, and waterfall charts for understanding cumulative positive and negative contributions to a total valu

IN-LAB EXERCISE:

OBJECTIVE:

To transform COVID-19 data into a structured format suitable for Business Intelligence (BI) reporting and design interactive dashboards with Key Performance Indicators (KPIs) using Tableau and Power BI.

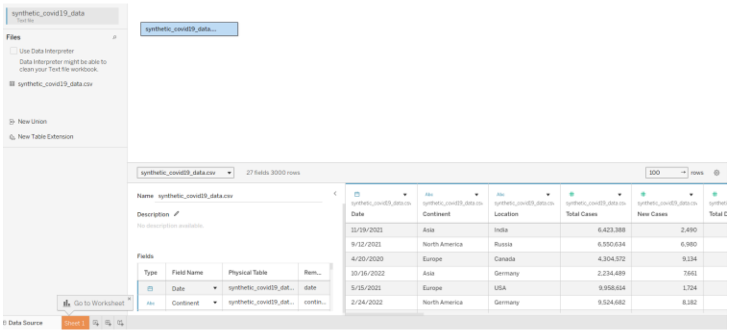
PROCEDURE:

1. Data Collection:

- Import the **COVID-19 dataset** from inbuilt sources like *Our World in Data* or CSV files containing country-level COVID statistics (e.g., new cases, deaths, vaccinations).
- Datasets may include fields such as *date*, *location*, *new_cases*, *total_cases*, *new_deaths*, *total_deaths*, *people_vaccinated*, etc.

Dataset Source: https://www.kaggle.com/datasets/khushikyad001/covid-19-global-dataset

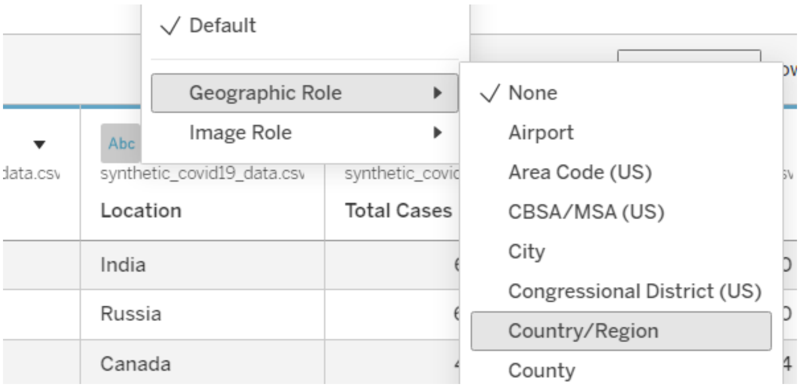
In Tableau:



2. Data Cleaning and Transformation:

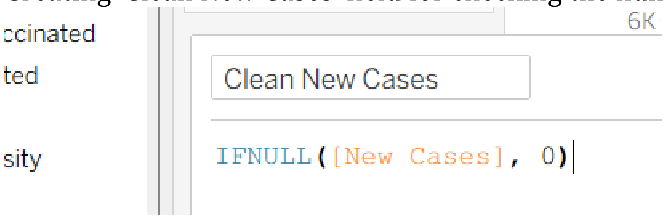
- Convert columns like *date* to proper date formats.

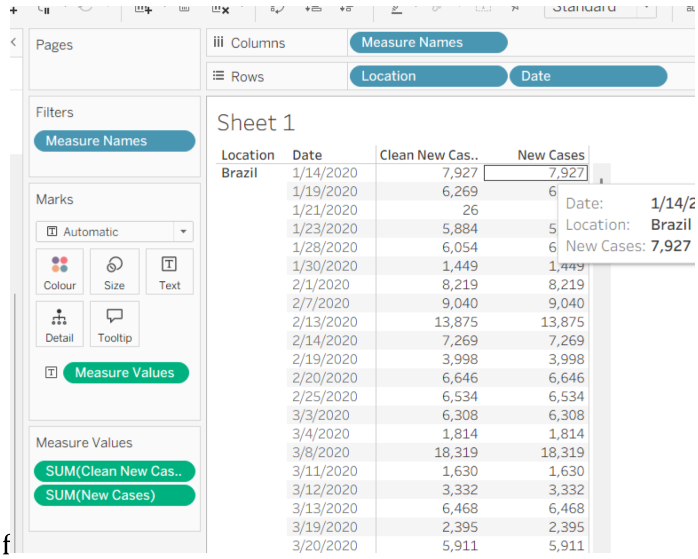
Formatting the Location Attribute as Country/Region:



- Handle missing or null values in *new_cases*, *total_cases*, etc.

Creating 'Clean New Cases' field for checking the number of missing values:

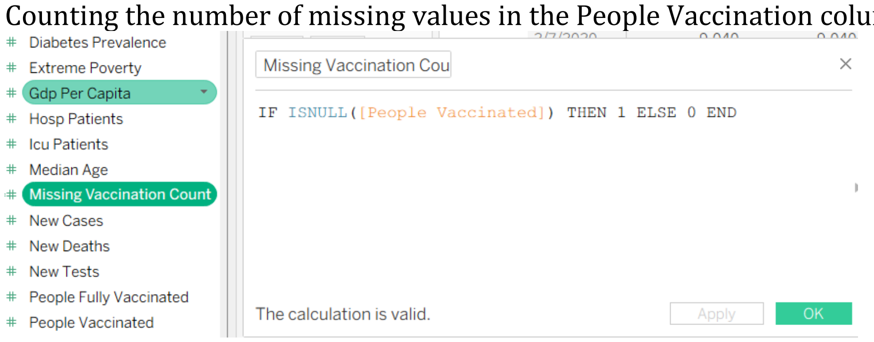




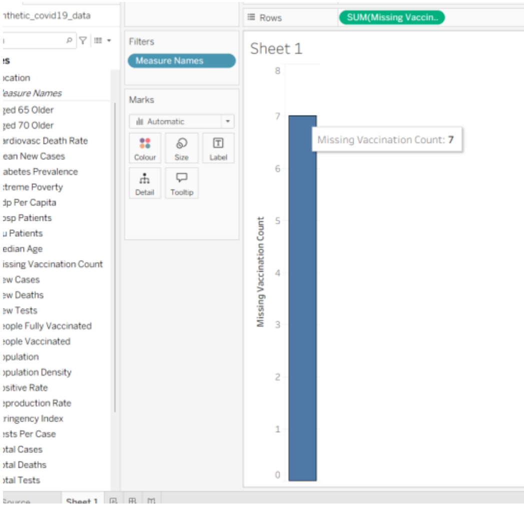
Location	Date	Clean New Cas..	New Cases
Brazil	1/14/2020	7,927	7,927
	1/19/2020	6,269	6
	1/21/2020	26	
	1/23/2020	5,884	5
	1/28/2020	6,054	6
	1/30/2020	1,449	1,449
	2/1/2020	8,219	8,219
	2/7/2020	9,040	9,040
	2/13/2020	13,875	13,875
	2/14/2020	7,269	7,269
	2/19/2020	3,998	3,998
	2/20/2020	6,646	6,646
	2/25/2020	6,534	6,534
	3/3/2020	6,308	6,308
	3/4/2020	1,814	1,814
	3/8/2020	18,319	18,319
	3/11/2020	1,630	1,630
	3/12/2020	3,332	3,332
	3/13/2020	6,468	6,468
	3/19/2020	2,395	2,395
	3/20/2020	5,911	5,911

From the above image, it is seen that there is no missing values in the New Cases

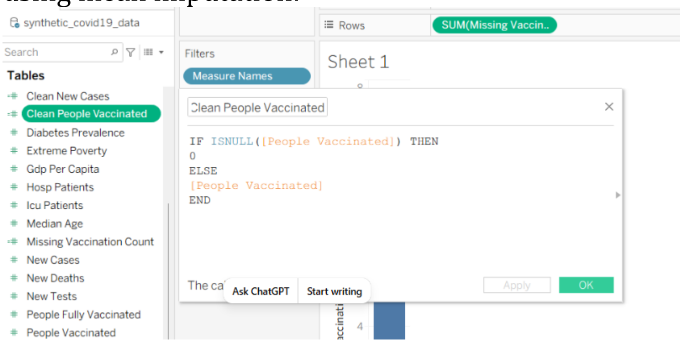
Counting the number of missing values in the People Vaccination column:



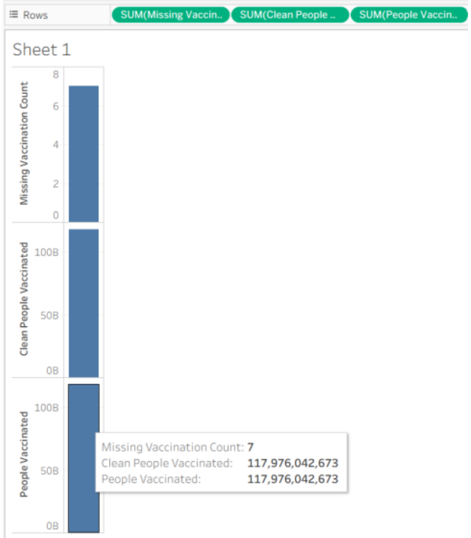
There are 7 missing samples in the People vaccinated column:



Handling missing values in the *People Vaccinated* column using a Tableau calculated field using mean imputation:

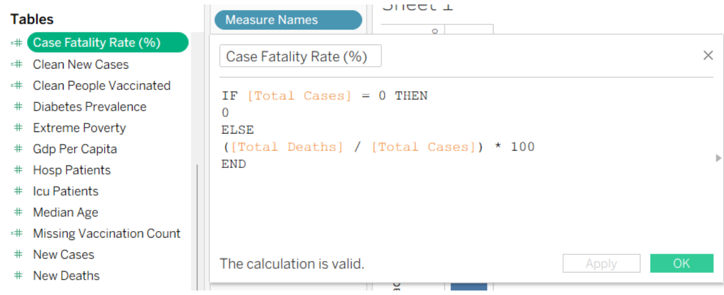


Now missing values are filled with mean:

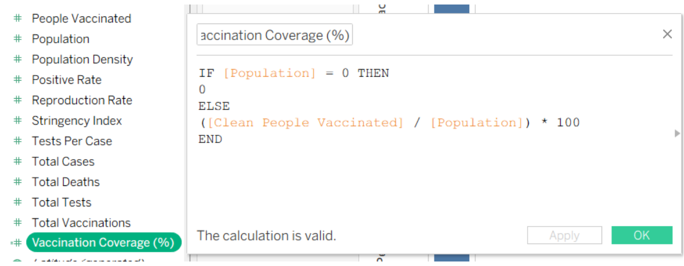


- Create calculated fields (e.g., daily case growth rate, case fatality rate, vaccination percentage).

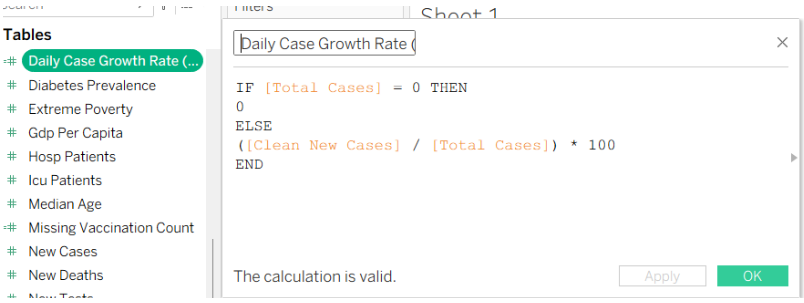
  Case fatality rate= (Total Deaths/Total Cases)*100:



Vaccination Coverage=(People Vaccinated/Population)*100



Daily Case Growth Rate=(New Cases/Total Cases)*100



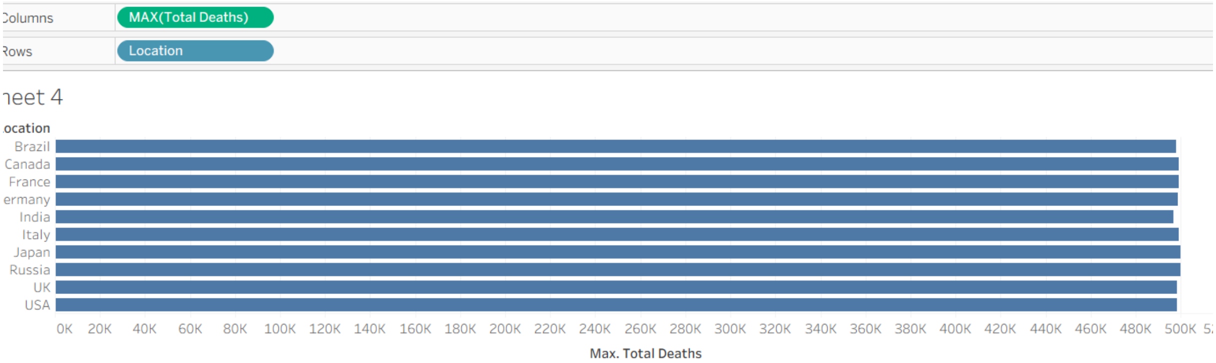
3. Data Aggregation:

- Aggregate data by *continent*, *country*, or *month*.

Aggregation of total new COVID-19 cases by continent using the SUM() aggregation function:

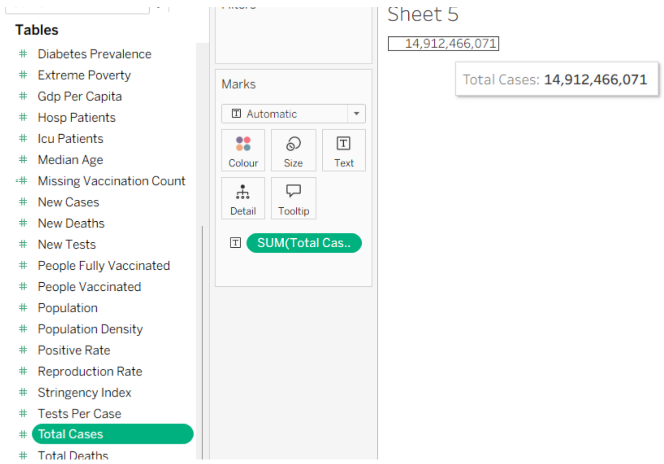
Average COVID-19 positivity rate by country using the AVG() aggregation function:

Maximum cumulative COVID-19 deaths by country using the MAX() aggregation function:

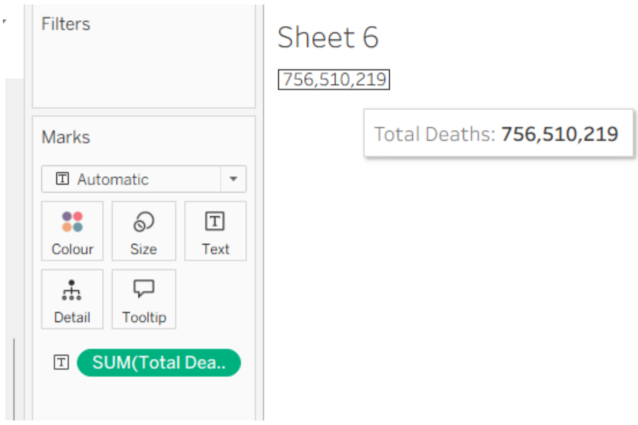


4. Designing KPIs:

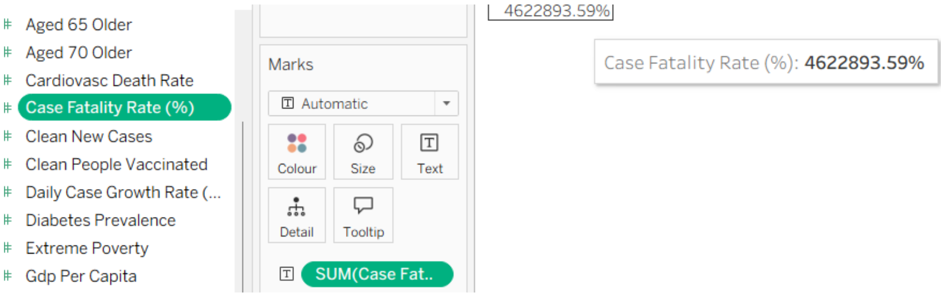
- Identify KPIs and visualize them:
    - Total Confirmed Cases – Created by dragging the Total Cases pill to the Text in the Marks Card:



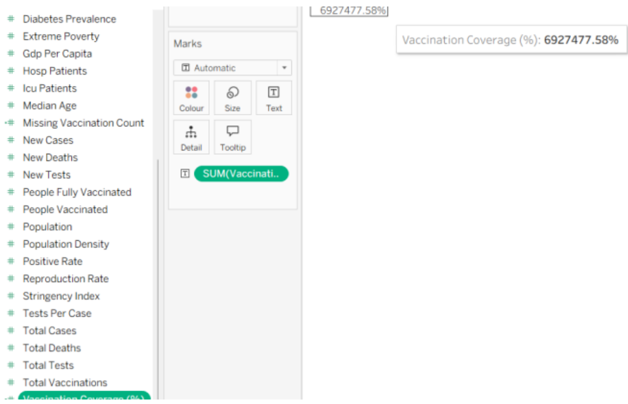
    - Total Deaths



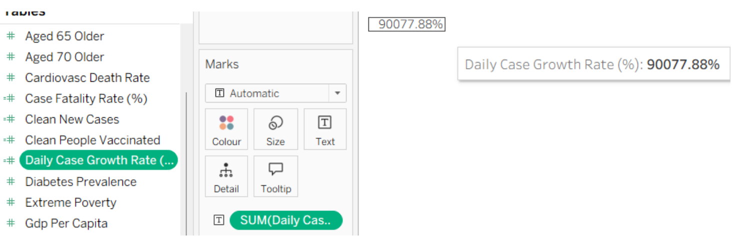
- Case Fatality Rate



- Vaccination Coverage (%)



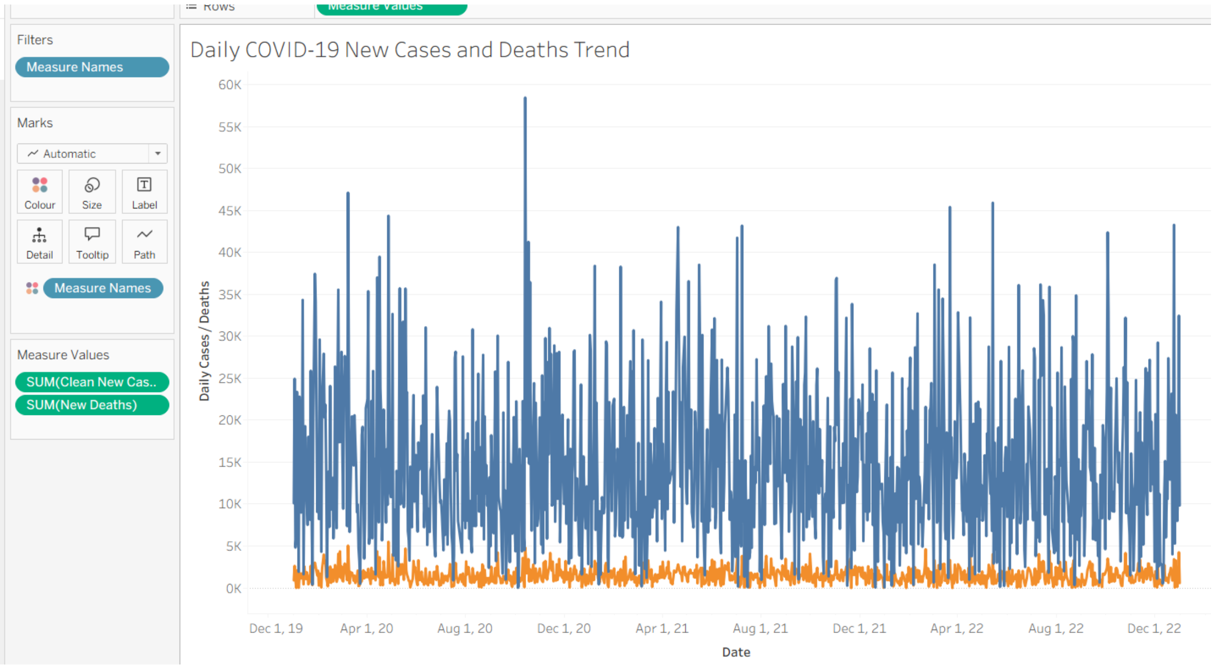
- Daily Case Growth Rate



5. Dashboard Development:

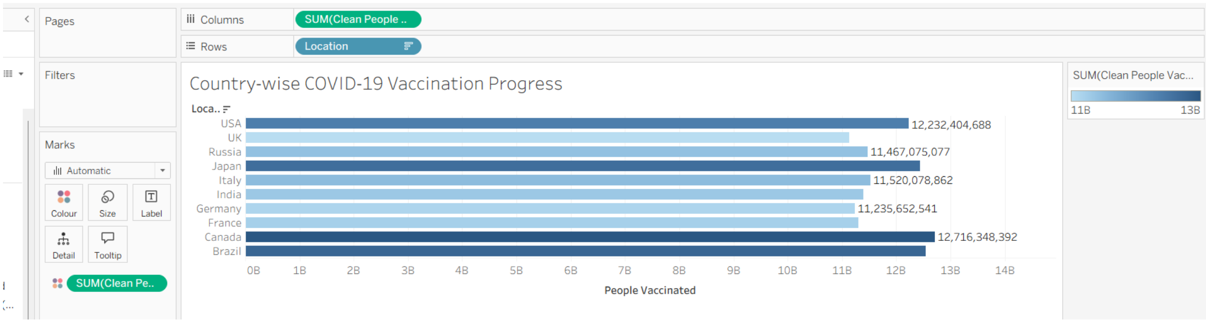
- Build visualizations such as:
  - Line charts for daily new cases and deaths

Line chart illustrating the daily trend of COVID-19 new cases and new deaths over time:



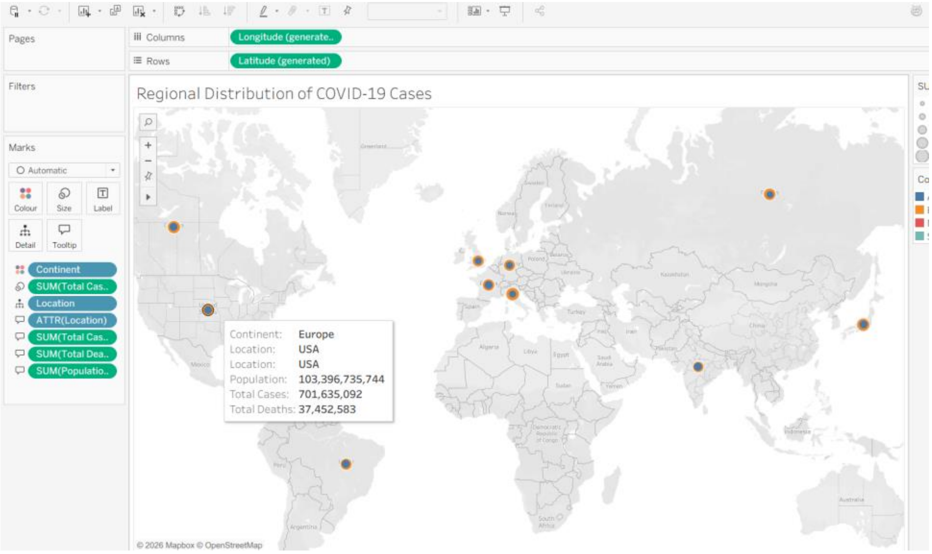
- Bar charts for country-wise vaccination progress

Horizontal bar chart showing country-wise COVID-19 vaccination progress based on number of people vaccinated:



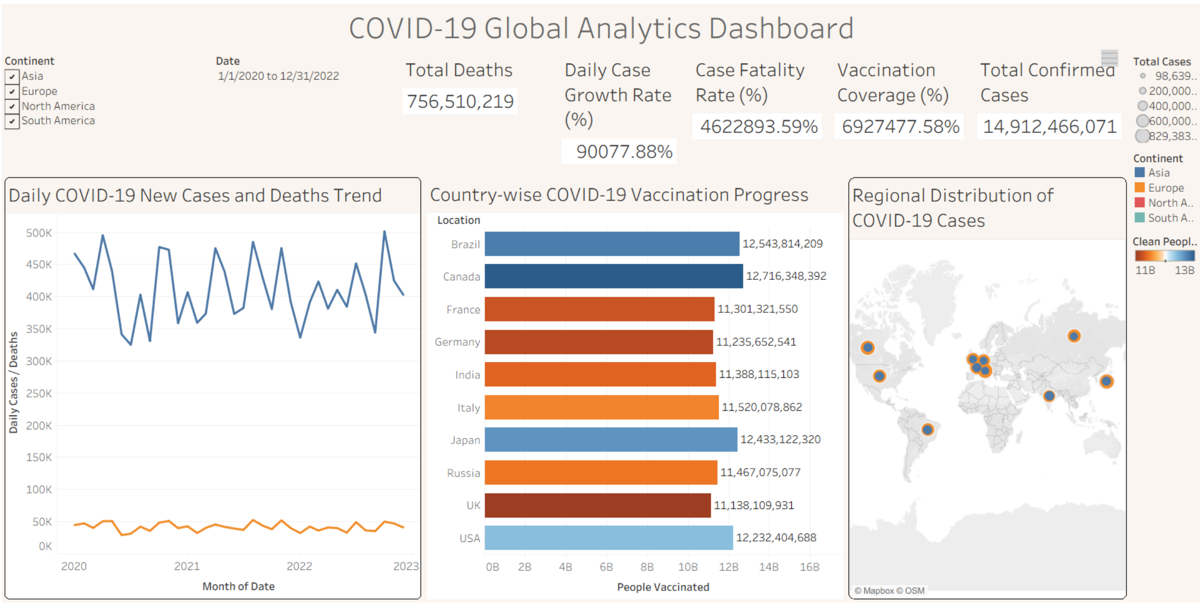
- Maps to show regional case distribution

Geographic distribution of COVID-19 cases by country using a Tableau map visualization:



- KPI cards for high-level insights

- Use filters (e.g., by continent, country) and slicers (e.g., by date range) for interactivity.



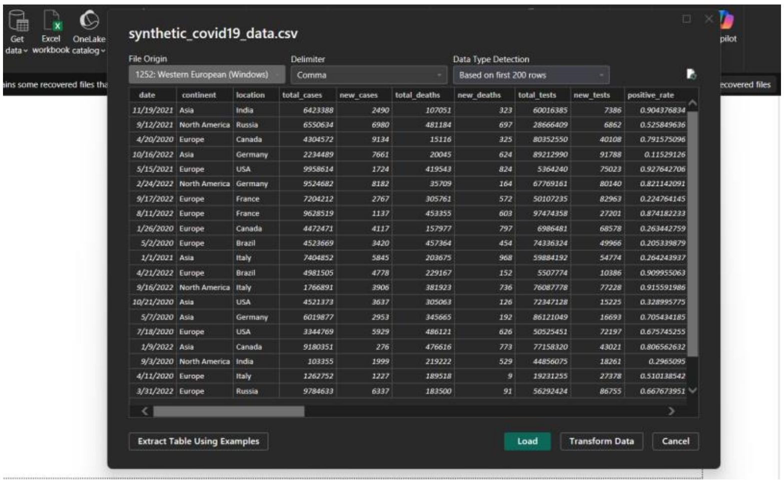
6. Publishing and Sharing:

- Publish the dashboard using Tableau Public/Server or Power BI Service.

- Share the dashboards with stakeholders through public or organizational portals.

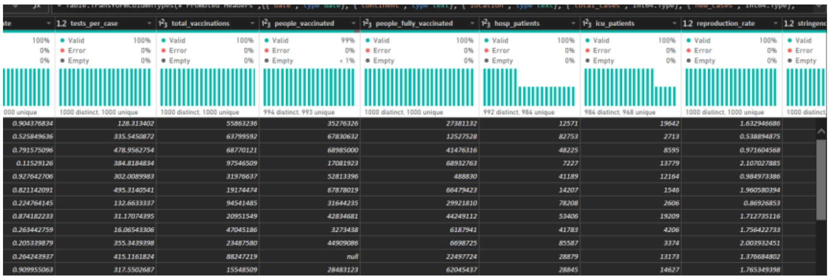
In PowerBI:

- **Load Dataset:**

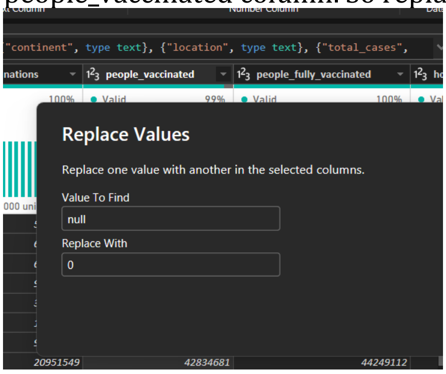


- Handle missing or null values in *new_cases*, *total_cases*, etc.

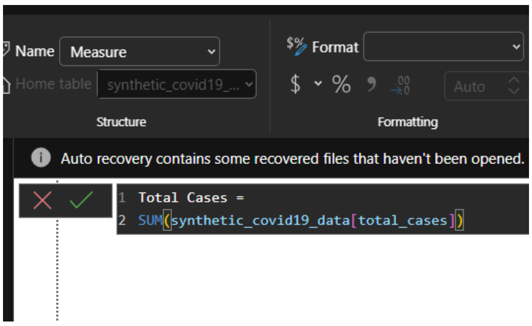
Creating 'Clean New Cases' field for checking the number of missing values:

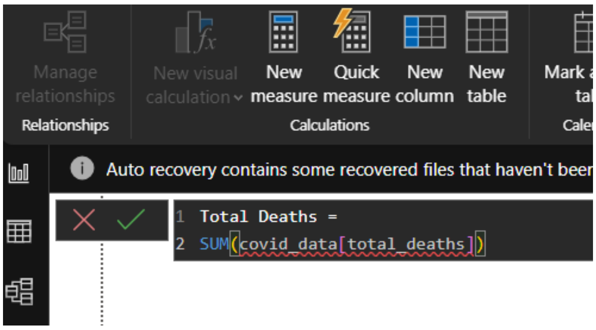


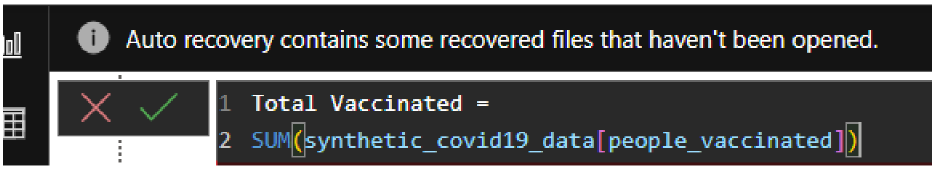
From PowerQuery Editor, we can see that there are some missing values in the people_vaccinated column. So replacing null values with 0:

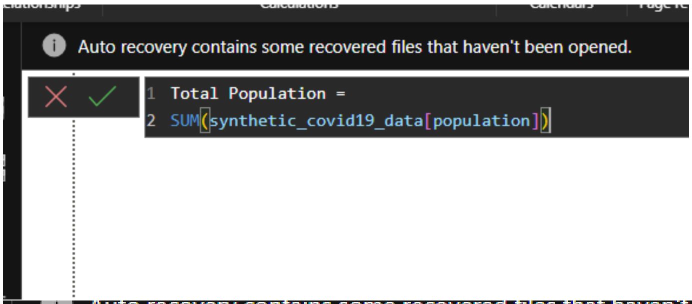


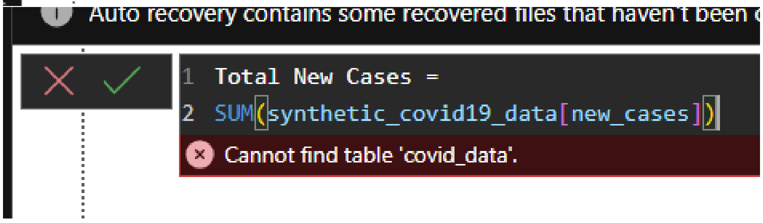
- Create calculated fields (e.g., daily case growth rate, case fatality rate, vaccination percentage).

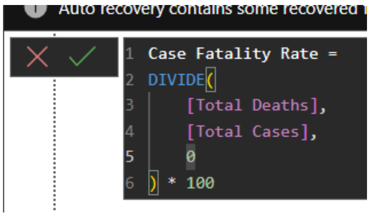


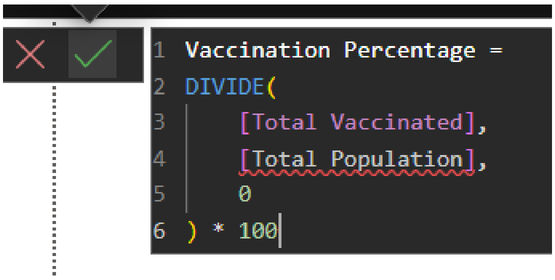


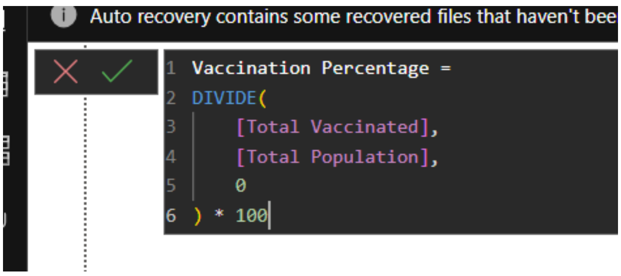


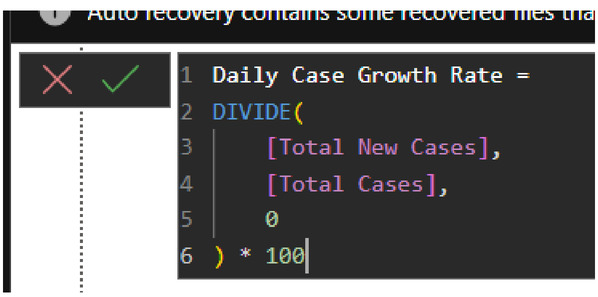




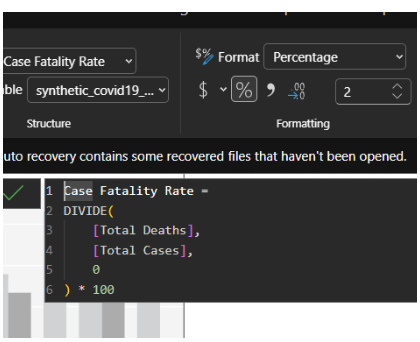


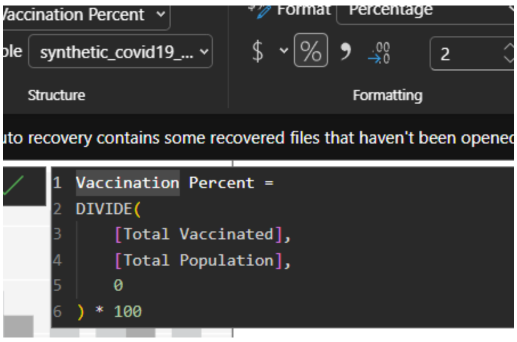


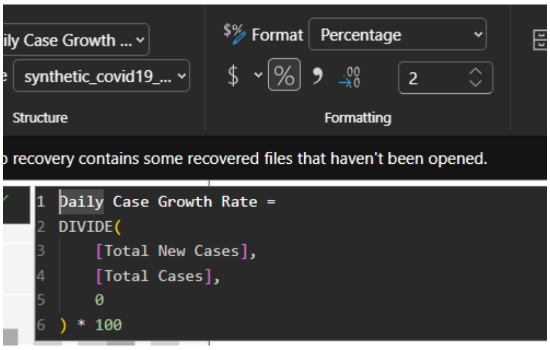




Formatting to 2 decimal places:



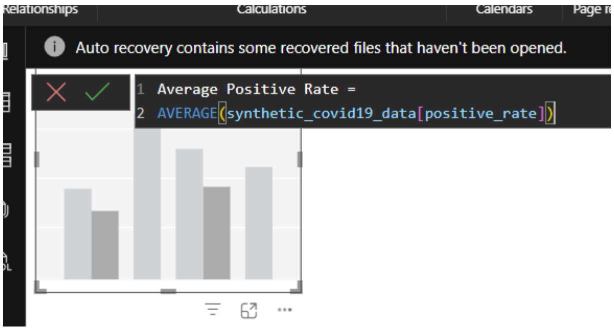


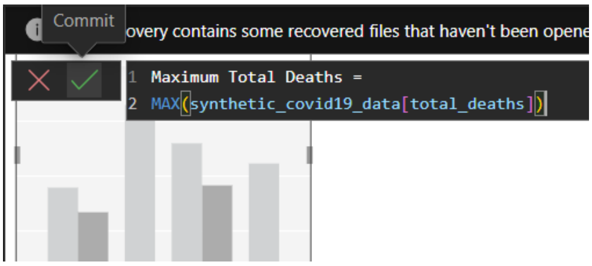


7. Data Aggregation:

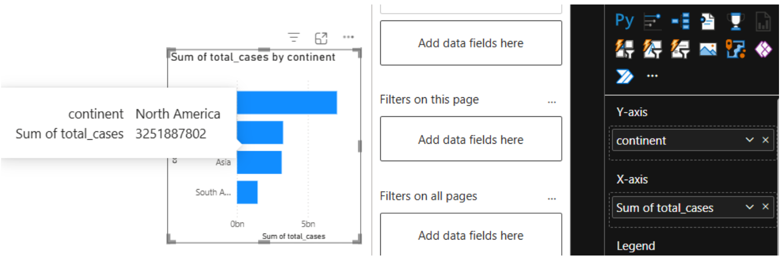
- Aggregate data by *continent*, *country*, or *month*.

Aggregation of total new COVID-19 cases by continent using the SUM() aggregation function:

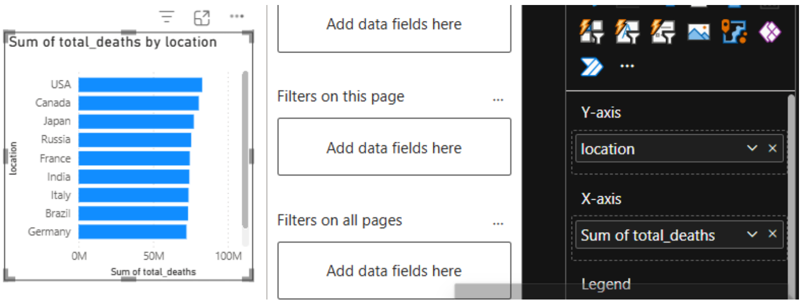


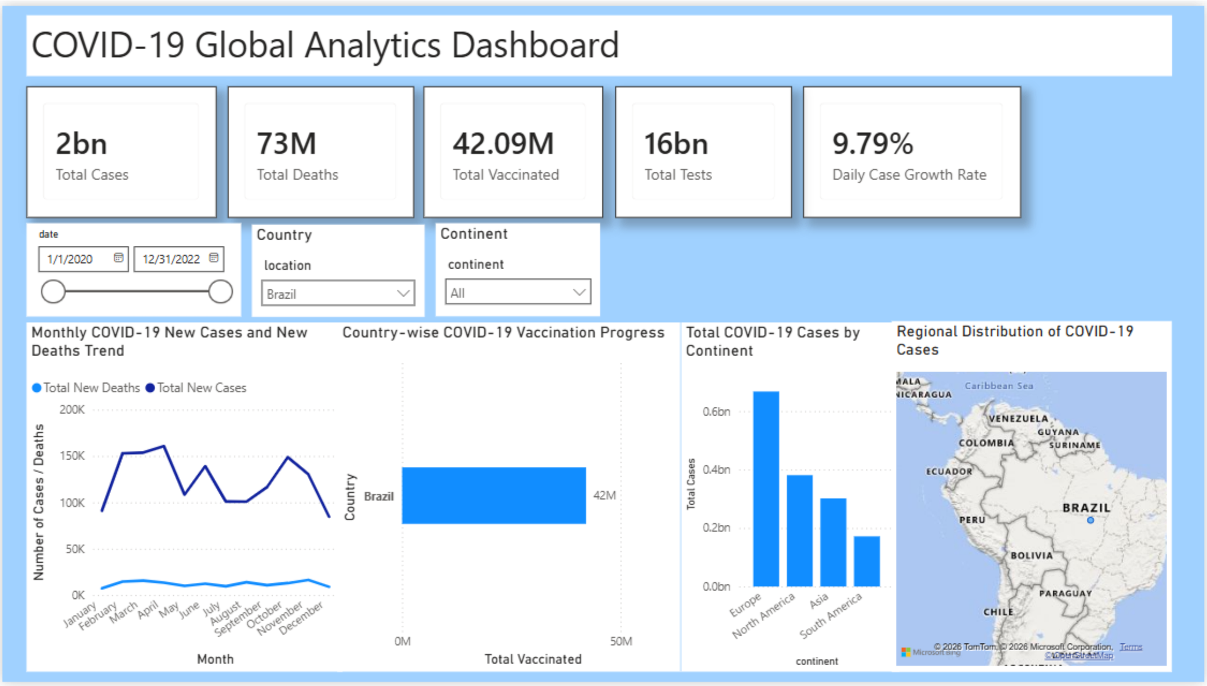


Demonstrating Aggregation by Continent:



Demonstrating Aggregation by Country:





8. Publishing and Sharing:

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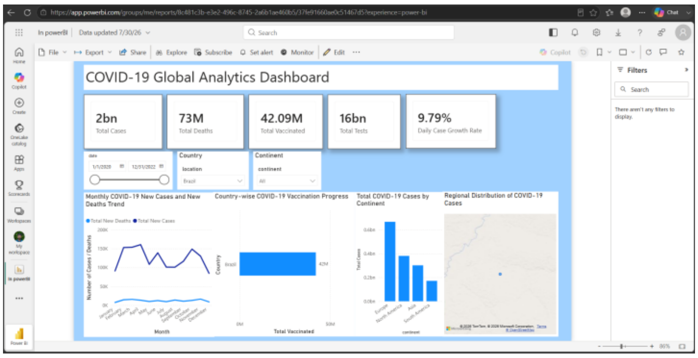
- Share the dashboards with stakeholders through public or organizational portals.



Published link:https://app.powerbi.com/groups/me/reports/8c481c3b-e3e2-496c-8745-2a6b1ae460b5/37fe91660ae0c51467d5?experience=power-bi

Experiment Dataset:

COVID-19 Global Dataset

Fields Used:

- **Date** – Date of the report
- **Location** – Country or region
- **New Cases** – Daily reported cases
- **New Deaths** – Daily reported deaths
- **People Vaccinated** – Cumulative vaccinated count
- **Total Cases** – Cumulative confirmed cases
- **Total Deaths** – Cumulative deaths
- **Population** – Total population of the country

Post Lab Questions:

1. How did data transformation improve the quality of insights in your COVID-19 dashboard?

Data transformation was an essential step in building the COVID-19 dashboard because it ensured that the data was accurate, consistent, and ready for analysis. Before creating the visualizations, missing values were handled, incorrect data types were corrected, and DAX measures were created to calculate important metrics. These transformations reduced the chances of calculation errors and improved the reliability of the dashboard.

As a result, the KPI cards and charts displayed meaningful insights, allowing users to analyze COVID-19 trends more effectively. A well-transformed dataset also made it easier to apply filters and generate interactive reports without inconsistencies.

2. What challenges did you encounter during the data cleaning process, and how did you resolve them?

While preparing the dataset, a few challenges were encountered that needed to be resolved before creating the dashboard.

- The people_vaccinated column contained missing (null) values, which could have affected calculations.
- These missing values were replaced with **0** using Power Query.
- The data types of numerical and date columns were verified to ensure accurate calculations.
- Since the dataset was synthetic, some percentage-based KPIs produced unrealistic values. To avoid misleading insights, only the most meaningful KPIs were included in the final dashboard.

These steps improved the quality of the dataset and ensured that the visualizations were generated without errors.

3. Which KPI provided the most valuable insight, and why?

Among all the KPI cards, Total Cases was the most valuable because it provides an immediate overview of the scale of the COVID-19 outbreak. It acts as the primary indicator for understanding the spread of the disease and serves as a reference for interpreting other metrics such as Total Deaths and Daily Case Growth Rate.

In addition, this KPI updates dynamically whenever users apply filters for country, continent, or date. This enables quick comparisons between different regions and helps users understand the overall trend of the pandemic more effectively.

4. How does interactivity in dashboards improve user experience?

Interactive dashboards make data exploration much easier because users can analyze information without changing the underlying report. Instead of viewing static charts, they can customize the dashboard according to their requirements.

For example, in this dashboard, slicers were added for Country, Continent, and Date. When a user selects a value from any slicer, all the KPI cards, charts, and the map update automatically. This allows users to focus on specific regions or time periods, making the dashboard more engaging, user-friendly, and helpful for decision-making.

5. What steps would you take to further enhance the dashboard's effectiveness?
Although the dashboard provides useful insights, there are several improvements that could make it even more effective.

- Replace the synthetic dataset with a real-time COVID-19 dataset.
- Add additional KPIs such as Recovery Rate, Active Cases, and Testing Coverage.
- Include drill-through pages to provide detailed country-level analysis.
- Improve the dashboard design using custom themes and a more consistent color palette.
- Add forecasting features to analyze future trends and support better decision-making.

Implementing these enhancements would make the dashboard more informative, interactive, and suitable for real-world business intelligence applications.

RESULT:

The COVID-19 dataset was successfully cleaned, aggregated, and transformed for visual analysis using Tableau and Power BI. Missing values, data type mismatches, and inconsistencies were addressed. KPIs such as vaccination rate and daily case growth offered valuable insights into the pandemic trends across countries and timeframes. The resulting interactive dashboards allowed for effective public health monitoring and communication.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	**5**	
In Lab Exercise	**10**	
Post Lab Exercise	**5**	
Viva	**10**	
Total	**30**	
Faculty Signature		